

Environmental Science and Sustainability

FYBBA SEMESTER- 1 SEC 101	Environmental Science and Sustainability	2L:0T:0P	2 Credits
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Course description:

This course aims to familiarize students with fundamental environmental concepts and their relevance to business operations, preparing them to address forthcoming sustainability challenges. It is designed to equip students with the knowledge and skills needed to make decisions that account for environmental consequences, fostering environmentally sensitive and responsible future managers. The course content is divided into four comprehensive units. Unit 1 introduces basic environmental principles, the man-environment relationship, and sustainability issues. Unit 2 focuses on ecosystems, biodiversity, and sustainable practices. Unit 3 addresses environmental pollution, waste management, and sustainable development strategies. Finally, Unit 4 explores social issues, environmental legislation, and practical applications through hands-on fieldwork. Through this holistic approach, students will gain a deep understanding of environmental processes, the importance of sustainable practices, and their role in promoting sustainability within business contexts.

Course Objective(s):

1. This course aims to familiarize students with basic environmental concepts, their relevance to business operations, and forthcoming sustainability challenges
2. This course will equip students to make decisions that consider environmental consequences.
3. This course will enable future business graduates to become environmentally sensitive and responsible managers.

Course Content:

Unit 1: Understanding Environment, Natural Resources, and Sustainability

Fundamental environmental concepts and their relevance to business operations;
Components and segments of the environment, the man-environment

relationship. Concept of sustainability;

Natural resources and associated problems, Non-renewable Resources, Renewable Resources - A) Forest resources B) Water resources, C) Mineral resources, D) Food resources, E) Energy resources F) Land resources, Role of an individual in conservation of natural resources

Unit 2: Ecosystems, Biodiversity, and Sustainable Practices

Concept of an ecosystem, Structure and function of an ecosystem, Producers, consumers and decomposers, Energy flow in the ecosystem-1.The water cycle, 2.The Carbon cycle, 3.The Oxygen cycle, 4, The Nitrogen cycle, 5.The energy cycle, 6.Integration of cycles in nature, Ecological succession, Food chains

The importance of biodiversity, the threats it faces, and the methods used for its conservation.

Unit 3: Environmental Pollution, Waste Management, and Sustainable Development

Definition, Causes, effects and control measures of - Air pollution, Water pollution, Soil pollution, Marine pollution, Noise pollution, Thermal pollution, Nuclear hazards

Solid waste Management: Causes, effects and control measures of urban and industrial wastes

Disaster management: floods, earthquake, cyclone and landslides. From Unsustainable to Sustainable development, Urban problems related to energy, Water conservation, rain water harvesting, water shed management

Unit 4: Social Issues, Legislation, and Practical Applications

Overview of key environmental legislation and the judiciary's role in environmental protection, including the Water (Prevention and Control of Pollution) Act of 1974, the Environment (Protection) Act of 1986, and the Air (Prevention and Control of Pollution) Act of 1981.

Readings:

Text Books (Latest Editions):

- Poonia, M.P. *Environmental Studies*, Khanna Book Publishing Co.
- Bharucha, E. *Textbook of Environmental Studies*, Orient Blackswan Private Ltd.
- Dave, D., & Katewa, S. S. *Text Book of Environmental Studies*. Cengage Learning India Pvt Ltd.
- Rajagopalan, R. *Environmental studies: from crisis to cure*, Oxford University Press.



- Miller, G.T. & Spoolman S. *Living in the Environment*. Cengage.
- Basu, M., & Xavier Savarimuthu, S. J. *Fundamentals of environmental studies*. Cambridge University Press.
- Roy, M. G. *Sustainable Development: Environment, Energy and Water Resources*. Ane Books.
- Pritwani, K *Sustainability of business in the context of environmental management*. CRC Press.
- Wright, R.T. & Boorse, D.F. *Environmental Science: Toward A Sustainable Future* (13th ed.). Pearson.

References

Web links:

- <https://www.ourplanet.com>
- <https://www.undp.org/content/undp/en/home/sustainable-development-goals.html>
- www.myfootprint.org
- <https://www.globalchange.umich.edu/globalchange1/current/lectures/king/ecosystem/ecosystem.html>

Course Outcome(s):

1. Explore the basic environmental concepts and issues relevant to the business and management field.
2. Recognize the interdependence between environmental processes and socio-economic dynamics.
3. Determine the role of business decisions, policies, and actions in minimizing environmental degradation.
4. Identify possible solutions to curb environmental problems caused by managerial actions.
5. Develop skills to address immediate environmental concerns through changes in business operations, policies, and decisions.

